Wireless Powering of Ionic Polymer Metal Composites Toward Hovering Microswimmers

Karl Abdelnour, Adam Stinchcombe, Maurizio Porfiri, *Member, IEEE*, Jun Zhang, and Stephen Childress

***Abstract*—In this paper, we present the design of a wireless powering system for ionic polymer metal composites (IPMCs). The system design is motivated by the need for enabling technologies to replicate hovering flight and swimming in biological systems. IPMC wireless powering is achieved by using radio frequency magnetically coupled coils and in-house designed power electronics for low-frequency IPMC actuation. Parameters of the circuit components describing the resonantly coupled coils and the IPMC are experimentally identified. The power transfer from the external power source to the receiver at the IPMC is experimentally analyzed for a broad range of system parameters. Flow visualization and particle image velocimetry are used to ascertain the system capabilities. Moreover, the IPMC vibration in the wireless and wired configurations is compared.**

***Index Terms*—Actuator, hovering, ionic polymer metal composite (IPMC), underwater robotics, wireless powering.**

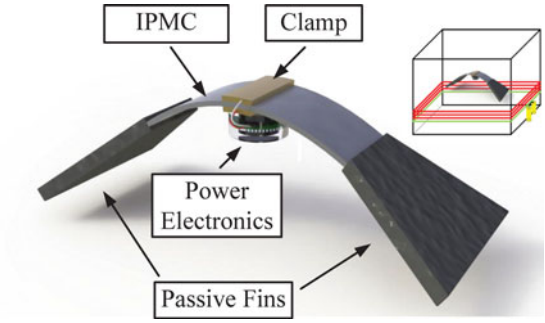


Fig. 1. Conceptual design of the robotic microswimmer.

I. Introduction

THE STUDY of hovering in insects, birds, and fish continues to garner considerable research focus within the fluid dynamics and robotics communities (see, for example, [1]–[8]). Experimental studies on passive bodies in the presence of oscillating fluid flows are reported in [9] and [10]. Robotic wings have been used to study the dynamics of lift production in [11]–[16]. Nevertheless, such systems do not allow for centimeter-scale analysis of free active hovering in placid environments, whose characterization is presently limited to numerical simulations [17]–[20]. The main objective of this paper is to develop enabling technologies toward the realization of miniature hovering robotic microswimmers.

Ionic polymer metal composites (IPMCs) are promising smart materials for constructing active surfaces due to their limited weight and low power consumption (see, for example, [21]–[24]). IPMCs are composed of an ionomeric membrane plated by noble metal electrodes and infused with an electrolytic solution. A voltage across the IPMC electrodes induces an electric field across its membrane that, in turn, promotes a reorganization of the charge and solvent distribution within the membrane. These phenomena are responsible for a macroscopic bending of the IPMC (see, for example, [23], [25]–[29]).

IPMCs find potential applications in the field of underwater robotics for their low activation voltage, silent operation, and flexibility. Biomimetic underwater propulsion based on IPMCs is demonstrated, for example, in [30]–[36] and the hydrodynamics induced by an IPMC vibrating in a quiescent fluid is experimentally studied in [37] and [38] and numerically explored in [39]. Available implementations of IPMCs as underwater propulsors use a wired or local power system, in which the power source resides onboard the vehicle. A large portion of the size and weight of these underwater vehicles is generally attributed to their batteries and onboard electronics (see, for example, [30] and [31]). This creates an obstacle for the development of actively hovering bodies.

The remarkable performance of IPMCs as underwater propulsors and the considerable dimensions of typical power sources serve as the main motivation for this study. An illustrative example of the proposed wirelessly powered robotic microswimmer design is shown in Fig. 1. The swimmer is intended to hover in a quiescent aqueous environment by wirelessly receiving power through magnetically coupled coils placed around a laboratory tank and connected to an external power source. The envisioned microswimmer consists of an IPMC with compliant passive fins, a mechanical clamp, and power electronics. The frequency and amplitude of the IPMC vibration are *a priori* determined through the selection of the constructive parameters based on the targeted application. Resonant coupling is selected as the wireless powering scheme for the robotic microswimmer due

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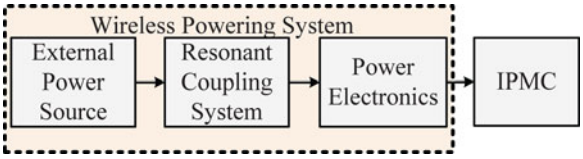


Fig. 2. Outline of the proof-of-concept platform.

to its superior performance as compared to inductive coupling, as discussed in [40]–[44]. Emerging applications of wireless powering technologies, spanning from multivehicle robotics to medical applications, can be found, for example, in [45]–[51].

In this paper, we present a proof-of-concept platform for IPMC wireless powering. More specifically, we design and implement a wireless powering system which is composed of an external power source, a resonant coupling system, and power electronics. The resonant coupling system is implemented using four radio frequency magnetically coupled coils. The power electronics are designed to transform the radio frequency wirelessly transmitted signal into a low-frequency square wave for IPMC powering. The resonant coupling system is modeled using linear circuit theory. The mutual inductances among the magnetically coupled coil pairs are experimentally identified. Additionally, we characterize the power electronics system by measuring the power output as a function of the load impedance for a broad range of shunting loads and we identify the electrical properties of the IPMC from its transient response under a step voltage input. Moreover, we elucidate the IPMC integration into the wireless powering system and compare the tip displacement of the IPMC in the wireless and wired configurations. Finally, we perform a flow visualization experiment and a particle image velocimetry (PIV) study on the wirelessly powered IPMC to illustrate the system capabilities.

II. System Description

A schematic of the proof-of-concept platform is shown in Fig. 2. The platform consists of: 1) the external power source which is designed to drive the resonant coupling system; 2) the resonant coupling system that allows for wireless power transfer at radio frequencies from the external power source to the receiver; 3) the power electronics circuit that modifies the frequency content of the wirelessly transmitted signal for IPMC actuation; and 4) the IPMC propulsor. Components 1)–4) define the wireless powering system.

A. External Power Source

The circuit diagram representing the design of the external power source is illustrated in Fig. 3. Here, a square-wave signal triggers an IRF510A MOSFET to shape the radio frequency input to the resonant coupling system. The system represents a low-cost solution suitable for rapid implementation.

In Fig. 3, V_{DC} is the voltage across a DC power supply which is set to $15\,\mathrm{V}$ and V_{T} represents the voltage of the square-wave triggering signal that varies in the range 0–$5\,\mathrm{V}$ and has an adjustable frequency. In addition, V_{s} is the output voltage of the external power source that is in turn delivered to the resonant coupling system.

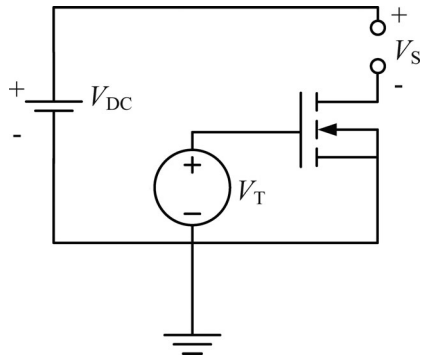


Fig. 3. Circuit diagram of the external power source.

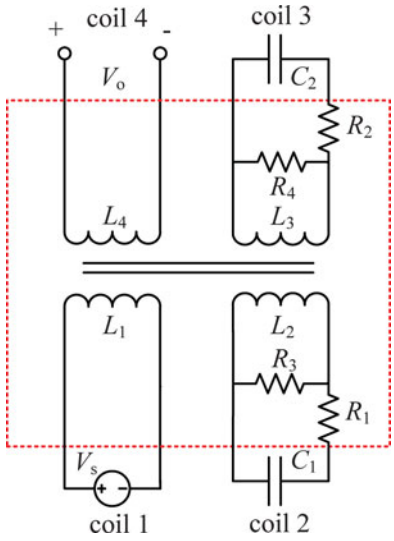


Fig. 4. Circuit diagram of the resonant coupling system; elements within the dashed lines describe the magnetic coupling.

B. Resonant Coupling System

The resonant coupling system follows the design in [41] and consists of two pairs of resonantly coupled coils, referred to as the source and the receiving coil pairs. The source pair encloses the environment in which the receiving coil pair, to be embedded in the microswimmer, is positioned as schematically illustrated in Fig. 1. Coils in each pair are coaxial and share the same surface area to maximize inductive coupling. Following [41], different winding numbers are selected within each pair to reduce parasitic resistances and enhance the coupling among the pairs. Resonant coupling between the pairs is achieved through tuned capacitors shunting the coils with higher winding number in the receiving and source pairs. The circuit diagram representing the resonant coupling system is displayed in Fig. 4, in which coils 1 and 2 form the source pair and coils 3 and 4 comprise the receiving pair. Coil 1 in the source pair is connected to the external power source, whose output voltage is V_{s}, consistently with Fig. 3. Coil 4 in the receiving coil pair

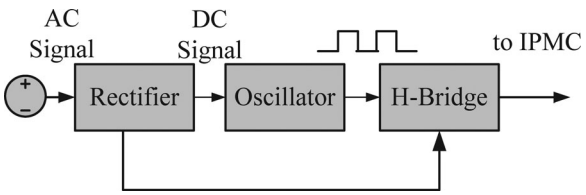


Fig. 5. Overview of the power electronics.

drives the power electronics and its voltage output is termed V_o. Both these coils have a minimal winding number to reduce the input/output resistance of the resonant coupling system. Coils 2 and 3 are resonantly coupled and internal resonance is achieved by selecting the shunting capacitances C_1 and C_2 so that

$$\frac{1}{\sqrt{L_2 C_1}} \approx \frac{1}{\sqrt{L_3 C_2}} \tag{1}$$

where L_2 and L_3 are the inductance of coils 2 and 3, respectively. In Fig. 4, resistances R_1, R_2, R_3, and R_4 account for the resistance of the resonantly coupled coils as well as further radio frequency energy losses.

The constants of the resonant coupling system are as follows. The number of turns in coils $1, ..., 4$ are 1, 21, 50, and 3, respectively. The diameters of the wires in coils $1, ..., 4$ are 3, 1.5, 0.5, and $0.25\,\mathrm{mm}$, respectively. The receiving coil pair has a circular cross section of $10\,\mathrm{mm}$ in diameter and is wound using a copper wire and insulated magnetic wire for coils 3 and 4, respectively. The source coil pair is wound using an insulated copper wire and placed around a transparent plastic tank of square cross section $161\,\mathrm{cm}^2$. In this proof-of-concept platform, the tank is used as a fixed mechanical support for the two coil pairs and the not yet water-proof power electronics; the IPMC is operated in a separate instrumented wet environment. The coil resistances are measured to be $R_1 = 0.9\,\Omega$ and $R_2 = 0.5\,\Omega$. The inductances of the coils are measured following [41] and are found to be $L_1 = 0.700\,\mu\mathrm{H}$, $L_2 = 157\,\mu\mathrm{H}$, $L_3 = 115\,\mu\mathrm{H}$, and $L_4 = 3.98\,\mu\mathrm{H}$. The measured capacitances of the ceramic capacitors used to tune the source coil and receiving coil pairs to a similar resonant frequency, in the range of 1.6–1.7 MHz, are $C_1 = 52\,\mathrm{pF}$ and $C_2 = 82\,\mathrm{pF}$. Resistances R_3 and R_4 are separately identified from the transfer function between a voltage input, in the frequency range $100\,\mathrm{kHz}$ to $10\,\mathrm{MHz}$, and the current through the coil. Least-squares error minimization yields 50 and $45\,\mathrm{k}\Omega$ for R_3 and R_4, respectively. The mutual inductances M_{ij}, $i \neq j = 1, ..., 4$, depend on the configuration of coils i and j, that is, their proximity and relative orientation. Numerical values for a representative assembly are reported in the following experiments.

C. Power Electronics

In underwater applications, IPMCs are typically actuated in the frequency range 0.1–10 Hz (see, for example, [30], and [31]); therefore, the power electronics are designed to step down the wirelessly transmitted radio frequency into a low-frequency input to the IPMC. A schematic of the power electronics is shown in Fig. 5.

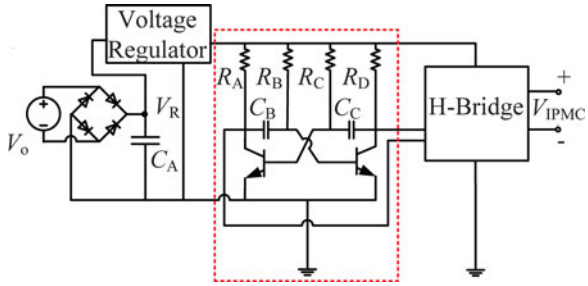


Fig. 6. Circuit diagram of the power electronics displaying the bridge rectifier and the voltage regulator (left), the ATO (enclosed by dashed lines), and the H-Bridge (right).

We use a bridge rectifier to convert the signal from AC to DC. An electrolytic reservoir capacitor $C_\mathrm{A} = 1000\,\mu\mathrm{F}$ at the output of the bridge rectifier filters the signal. The DC signal is delivered to a 5 V Fairchild 7805A voltage regulator for signal conditioning and then to an astable transistor oscillator (ATO) that generates a square wave. The ATO is a passive element oscillator that is selected for its small size and ease of implementation. The low current output of the ATO is used as an input to an ISL83202 H-Bridge that drives the IPMC and provides signal buffering. The H-Bridge is also powered by the conditioned DC signal from the output of the voltage regulator.

Fig. 6 shows the constituent elements of the power electronics circuit. Resistors R_A and R_D act as pull-up resistors and both have a nominal value of $220\,\Omega$. The oscillation frequency is given by the expression $[\ln(2) \cdot (R_\mathrm{B} C_\mathrm{B} + R_\mathrm{C} C_\mathrm{C})]^{-1}$ (see, for example, [52]). We select the nominal resistances of the resistors R_B and R_C to be $10\,\mathrm{k}\Omega$ and the nominal capacitances of the electrolytic capacitors C_B and C_C to be $100\,\mu\mathrm{F}$. This selection yields a nominal frequency of $720\,\mathrm{mHz}$ and maintains a duty cycle of 50%. In addition, V_R represents the rectified DC voltage available to the ATO and V_{IPMC} refers to the voltage across the IPMC.

D. IPMC Propulsor

The IPMC strip was procured from Environmental Robots (www.environmental-robots.com) and comprises a Nafion117 polymer core plated with platinum electrodes. The IPMC sample is stored in a 1 N solution of NaCl for approximately 2 h prior to an experimental campaign. In between usage trials, the IPMC is stored in deionized water. We clamp the IPMC strip between two aluminum elements for structural support. The clamp also serves as the electrical connection to the IPMC electrodes. The IPMC has a total length of $12\,\mathrm{mm}$, a width of $5.8\,\mathrm{mm}$, and a nominal thickness of $200\,\mu\mathrm{m}$. A passive compliant fin made of polyethylene is attached to the IPMC so that the propulsor has a free length of $40\,\mathrm{mm}$ and a width of $15\,\mathrm{mm}$ (see Fig. 7). During experiments, the IPMC and the clamp are fully immersed in deionized water.

III. Experiments

We conduct several experiments to identify the parameters and assess the performance of the proof-of-concept platform.

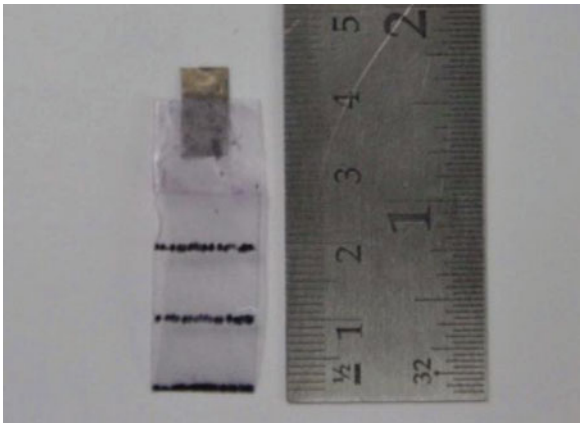


Fig. 7. IPMC propulsor, including IPMC strip and passive fin; scale is in centimeters.

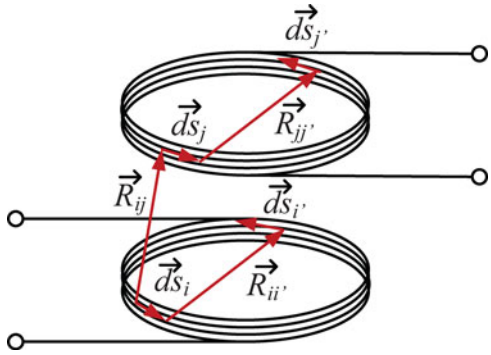


Fig. 8. Schematic of two mutually coupled inductors.

Experimental identification encompasses estimating the mutual inductances M_{ij}'s of the resonant coupling system in Fig. 4 and the parameters of a lumped IPMC electrical model. The system performance is ascertained through the measurement of the wireless power transfer and the motion of the IPMC propulsor underwater. In addition, a flow visualization experiment and a PIV study are conducted to illustrate the motion imparted by the IPMC propulsor to the surrounding aqueous environment and quantify its thrust production.

A. Parameter Identification of the Resonant Coupling System

In the frequency domain, the circuit in Fig. 4 is described by the following set of equation:

$$\tilde{V}_{\mathrm{s}} = j2\pi f\tilde{\Gamma}_1 \tag{2a}$$

$$0 = R_3\frac{R_1+\frac{1}{j2\pi fC_1}}{R_3+R_1+\frac{1}{j2\pi fC_1}}\tilde{I}_2 + j2\pi f\tilde{\Gamma}_2 \tag{2b}$$

$$0 = R_4\frac{R_2+\frac{1}{j2\pi fC_2}}{R_4+R_2+\frac{1}{j2\pi fC_2}}\tilde{I}_3 + j2\pi f\tilde{\Gamma}_2 \tag{2c}$$

$$\tilde{V}_{\mathrm{o}} = j2\pi f\tilde{\Gamma}_4. \tag{2d}$$

Here, f is the frequency of excitation, a superimposed tilde is used to identify Fourier transformed quantities, and the flux linkages Γ_1, ..., Γ_4 are related to the corresponding currents I_1, ..., I_4 by

$$\begin{bmatrix}\Gamma_1\\ \Gamma_2\\ \Gamma_3\\ \Gamma_4\end{bmatrix} = \begin{bmatrix}L_1 & M_{12} & M_{13} & M_{14}\\ M_{12} & L_2 & M_{23} & M_{24}\\ M_{13} & M_{23} & L_3 & M_{34}\\ M_{14} & M_{24} & M_{34} & L_4\end{bmatrix}\begin{bmatrix}I_1\\ I_2\\ I_3\\ I_4\end{bmatrix}. \tag{3}$$

A dimensionless measure of the mutual coupling between the ith and jth coils is represented by $\kappa_{ij} = M_{ij}/\sqrt{L_iL_j}$. The coupling coefficients are identified by comparing predictions from the model with experimental measurements of the transfer function $\tilde{V}_{\mathrm{o}}/\tilde{V}_{\mathrm{s}}$ for a broad range of frequencies under open circuit condition at coil 4. Identification is performed by using nonlinear least-squares error minimization (see, for example, [53]). An initial guess for the system parameters is obtained through idealized estimates from the software package MandK [54] that directly evaluates Neumann integrals as a function of the coils' mutual configurations. For thin filaments, this corresponds to

$$M_{ij} = \frac{\mu_0}{4\pi}\oint_{\gamma_i}\oint_{\gamma_j}\frac{\vec{ds}_i\cdot\vec{ds}_j}{|\vec{R}_{ij}|} \tag{4}$$

Here, γ_i and γ_j denote the curves spanned by the coils, $\vec{R}_{ij}$ is the displacement between two points on the coils, while $\vec{ds}_i$ and $\vec{ds}_j$ denote vectors tangent to coils i and j, respectively, and μ_0 is the magnetic constant. The computation of the inductances $L_1, \ldots, L_4$ requires adapting equation (4) to account for inherent singularities. Fig. 8 illustrates the notation used for two mutually coupled coils.

The values of the coupling coefficients are $\kappa_{12} = 0.330$, $\kappa_{13} = 0.030$, $\kappa_{14} = 0.012$, $\kappa_{23} = 0.050$, $\kappa_{24} = 0.011$, and $\kappa_{34} = 0.500$ when the receiving coil pair is placed in the center of the plane of the source coil pair with the same orientation. Fig. 9 shows the theoretical transfer function and the experimental measurements. It is seen that the resonant coupling system is well described by the circuit model in Fig. 4 using experimentally measured values for the system parameters and selecting appropriate values for the coupling coefficients. Fig. 9 also shows that the system is highly resonant in the vicinity of 1.7 MHz, yet two distinct peaks are visible due to parameter mismatch in coils 2 and 3 as per the computation of the resonance frequencies in (1).

B. Power Delivery

In order to determine the power output of the wireless powering system under different loading conditions, a resistor R and a capacitor C are placed in series at the output of the power electronics. Experiments are conducted by wirelessly driving the power electronics and selecting resistance and capacitance values in a range typical of IPMCs (see, for example, [55]). The receiving coil pair is placed in the center of the plane of the source coil pair with the same orientation. More specifically, resistances within the range of 10–400 Ω in increments of 10 Ω are taken, while seven electrolytic capacitors with capacitance values between 1370 and 14100 μF are used. For each combination of resistance and capacitance, the rms voltage across the resistor V_{rms} is measured. The average power delivered to the

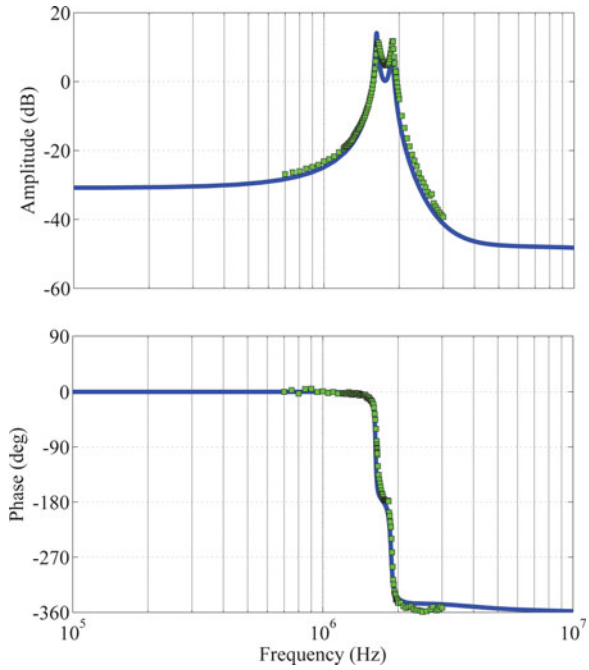


Fig. 9. Experimental (markers) and theoretical transfer functions $\tilde{V}_o/\tilde{V}_s$ (solid line) when the receiving coil pair is located in the center of the plane of the source coil pair with the same orientation.

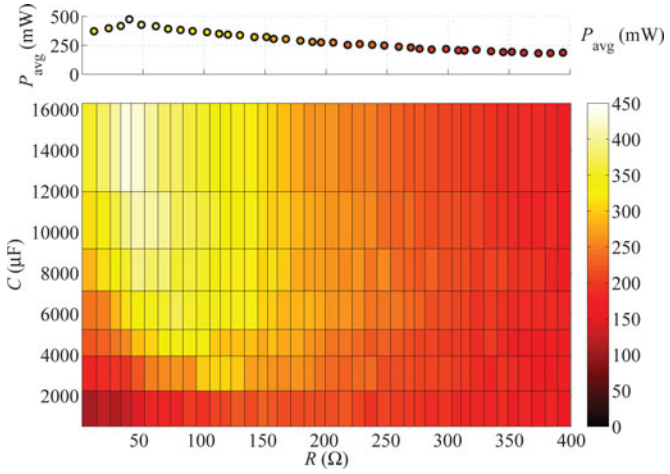


Fig. 10. Experimental data for the output power as a function of the load resistance and capacitance at the output of the power electronics. The top panel shows the power output for a purely resistive load.

load is estimated through

$$P_{\text{avg}} = \frac{V_{\text{rms}}^2}{R} \tag{5}$$

by recalling that a capacitor is a purely reactive element.

As expected, Fig. 10 demonstrates the existence of a maximum in P_{avg} due to impedance matching between the load and the wireless powering system. The dependence of P_{avg} on R decreases as C decreases. Hence, for relatively small capacitances, tailoring a value of R to maximize P_{avg} is not as critical as for large capacitive loads. We note that the reliability of predictions shown in Fig. 10 is based on the assumption that the electrical

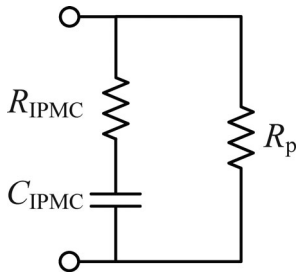


Fig. 11. Circuit model of the IPMC.

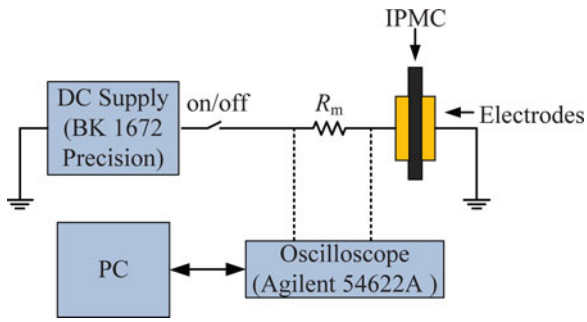


Fig. 12. Experimental setup for IPMC parameter identification.

response of the IPMC can be described through a two-terminal linear time-invariant network.

We comment that planar motion of the receiving coil has a minimal effect on the wireless power transfer, within 1%, whereas more dramatic losses can be experienced for out-of-plane motions or significant tilting of the receiving coil pair.

C. IPMC Electrical Properties

The IPMC electrical response is described by using a circuit model consisting of parallel connection of two branches, as shown in Fig. 11. The left branch describes the transient dynamics of the IPMC with a resistive element R_{IPMC} and a capacitive element C_{IPMC}. The right branch accounts for the DC resistance R_{p} of the IPMC which dominates its steady-state response. These parameters are taken as independent of the voltage level, thus yielding an elementary linear model. More comprehensive nonlinear models of IPMC electrical response are studied in [56]–[58]. The charge stored in the capacitor is proportional to the produced actuation (see, for example, [59]), and thus controls the IPMC overall deformation. In turn, such deformation has a minimal effect on the IPMC electrical behavior thus allowing for neglecting bidirectional actuator–sensor coupling (see, for example, [60]).

A schematic of the experiment used to identify the parameters of the model is shown in Fig. 12. A step voltage V_{step} is applied to the IPMC and the current through the IPMC is measured by recording the voltage across a $10\,\Omega$ series resistor R_{m}.

The linear circuit model describing the experimental setup and the IPMC equivalent circuit are represented by the

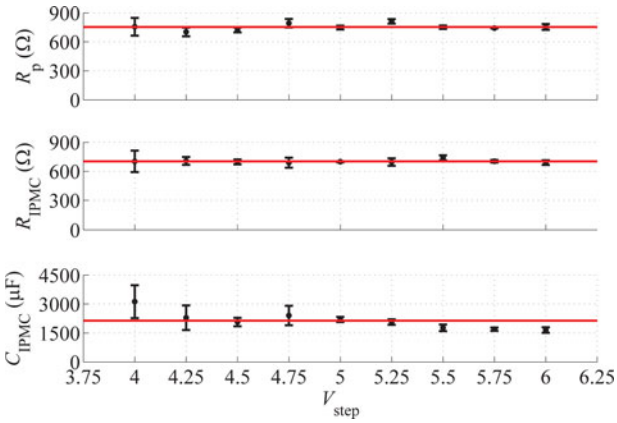


Fig. 13. Parameters $R_{\rm p}$, $R_{\rm IPMC}$, and $C_{\rm IPMC}$ versus step voltage input. Solid lines represent average parameter values. Error bars refer to one standard deviation computed on a set of five repeated measurements.

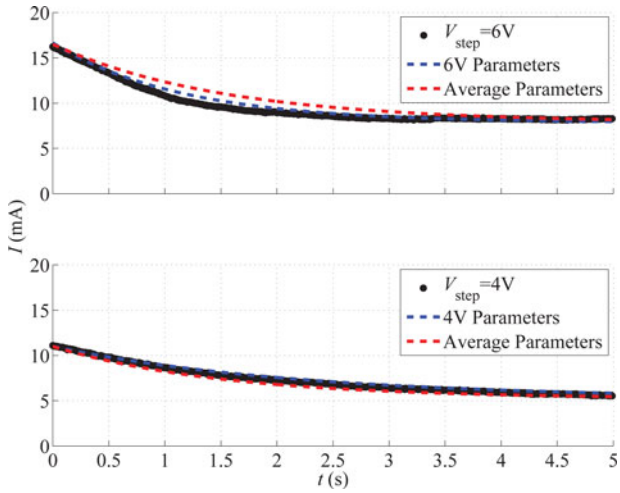


Fig. 14. Response of an IPMC sample to step voltage inputs of 6 and 4 V. The experimentally measured current is plotted with the theoretical curve obtained from (7) for the average parameters, as well as the parameters corresponding to a 6 and 4 V step, respectively.

differential equation

$$V_{\rm step} = C_{\rm IPMC}\left[R_{\rm IPMC}\left(1+\frac{R_{\rm m}}{R_{\rm p}}\right)+R_{\rm m}\right]\dot{V}_{\rm C}(t) + \left(1+\frac{R_{\rm m}}{R_{\rm p}}\right)V_{\rm C}(t) \quad (6)$$

where $V_{\rm C}$ is the voltage across the capacitor $C_{\rm IPMC}$. The IPMC is assumed to be initially fully discharged giving an initial condition $V_{\rm C}(0)=0$. The current through the IPMC is found by integrating (6), that is,

$$I(t) = I_{\rm ss} + \left[\frac{V_{\rm step}}{\frac{R_{\rm IPMC}R_{\rm p}}{R_{\rm IPMC}+R_{\rm p}}+R_{\rm m}} - I_{\rm ss}\right] \times \exp\left(-\frac{t}{C_{\rm IPMC}\left(R_{\rm IPMC}+\frac{R_{\rm m}R_{\rm p}}{R_{\rm m}+R_{\rm p}}\right)}\right) \quad (7)$$

where $I_{\rm ss} = V_{\rm step}/(R_{\rm p}+R_{\rm m})$ is the steady-state current.

The parameters from the equivalent circuit of the IPMC may be identified from an experiment for a single-step voltage. However, in order to assess the system linearity, a range of step voltages is used; namely, step voltages between 4 and 6 V in increments of 0.25 V are considered. For each step voltage, the parameters are obtained through comparison between the experimentally measured current and (7). The parameter $R_{\rm p}$ is computed directly from the steady-state current. The parameters $R_{\rm IPMC}$ and $C_{\rm IPMC}$ are obtained from least-squares fitting $I(t)-I_{\rm ss}$ against experimental data shifted by the steady-state current.

Fig. 13 shows the model parameter values for each step voltage. The DC resistance and the IPMC resistance remain approximately constant over this range of step voltages. The IPMC capacitance decreases slightly with increasing step voltage, yet clear trends may be difficult to isolate due to moderately large standard deviations. The averages of the parameter values over the nine step voltages are found to be 775 Ω, 703 Ω, and 2120 μF for $R_{\rm p}$, $R_{\rm IPMC}$, and $C_{\rm IPMC}$ with standard deviations of 50 Ω, 48 Ω, and 570 μF, respectively.

Fig. 14 displays the experimental time trace for current through the IPMC for the largest and smallest step voltages. Also shown is the linear model's prediction for the selected values from Fig. 13 against model predictions using the average parameters. This shows that the model is in good agreement with experimental findings in the entire range of step voltages considered in this study. We comment that more accurate fits can be obtained by using models incorporating multiple time scales (see, for example, [55] and [61]).

The IPMC circuit model can be integrated with the design chart in Fig. 10 to predict the average power delivered to the IPMC. Since the IPMC circuit model shown in Fig. 11 differs from the resistor–capacitor circuit used to obtain Fig. 10, we express an equivalent resistance $R_{\rm eq}$ and an equivalent capacitance $C_{\rm eq}$ for the model in Fig. 11 through the equivalent IPMC impedance $Z_{\rm eq}$ computed at the nominal frequency of the actuation square-wave signal $f_{\rm a}$ of 720 mHz, corresponding the expected fundamental harmonic component. In other words, we set $\mathrm{Re}[Z_{\rm eq}] = R_{\rm eq}$ and $\mathrm{Im}[Z_{\rm eq}] = -1/(2\pi f_{\rm a} C_{\rm eq})$ where $Z_{\rm eq} = (1/R_{\rm p} + 1/(R_{\rm IPMC} + 1/(j2\pi f_{\rm a} C_{\rm IPMC})))^{-1}$. We find that $R_{\rm eq} = 366\,\Omega$ and $C_{\rm eq} = 7960\,\mu$F, which predicts a power delivery of approximately 198 mW as per Fig. 10.

D. IPMC Motion Analysis

The components of the proof-of-concept platform are integrated into the experimental setup illustrated in Fig. 15. The wireless powering system drives the IPMC, which resides in a transparent plastic tank of approximately cubic shape and volume of 2500 cm^3 that is filled with deionized water. The tank is well illuminated and the IPMC motion is filmed by a camera. A digital oscilloscope records the DC voltage in the power electronics $V_{\rm R}$ and the voltage across the IPMC $V_{\rm IPMC}$ as per the terminology introduced in Fig. 6. The software suite Xcitex ProAnalyst is utilized to extract the time trace of the deflection δ along the length of the IPMC propulsor from the image

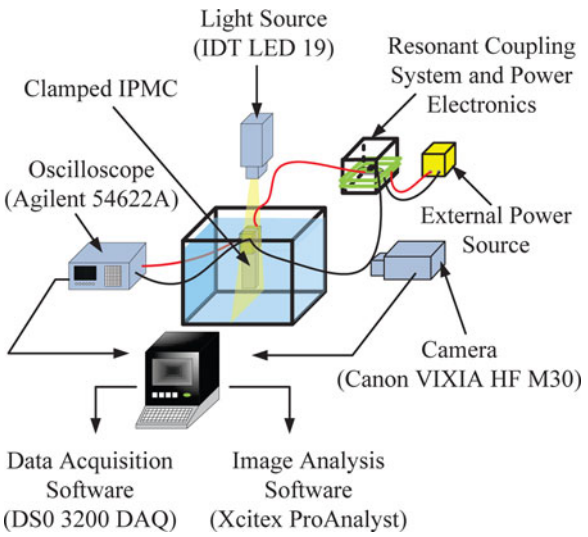


Fig. 15. Experimental setup for the IPMC motion analysis.

sequence. The deflection is measured with respect to the neutral position of the IPMC propulsor and the effect of axial displacement is neglected. The receiving coil pair is placed in the center of the plane of the source coil pair with the same orientation duplicating the arrangement used for the experimental results in Fig. 9.

Snapshots of the propulsor motion in Fig. 16 show that the vibration period is approximately $1.75\,\mathrm{s}$ and that the propulsor oscillates dominantly on its fundamental structural mode. In this time window, the propulsor is at rest at $0.00\,\mathrm{s}$ and attains its rightmost deflection at $0.50\,\mathrm{s}$, after which it returns through rest to its leftmost deflection at $1.5\,\mathrm{s}$. The IPMC sample utilized in this study has a mildly warped shape of the IPMC propulsor, as evidenced by Fig. 16 that may possibly amplified by buoyancy effects.

Displayed in Fig. 17 are the deflections at half, three quarters, and the full length of the IPMC propulsor. The quantities oscillate with the same phase, confirming that the propulsor vibrates along its fundamental mode. The ratios of these deflections are consistent with the lowest *in vacuo* structural mode of a homogenous beam (see, for example, [62]). In particular, the ratios of the deflection at three quarters length and half length to the tip deflection for the first mode shape are known to be approximately 0.66 and 0.34, respectively, which is consistent with the time traces in Fig. 17.

The propulsor tip deflection reported in Fig. 16 and Fig. 17 is comparable with typical underwater vibration induced by local power sources (see, for example, [30], and [31]). To better elucidate differences between wireless IPMC powering and traditional tethered configurations, we also consider the case in which the power electronics are connected directly to a DC power supply. This assembly is referred to as the wired configuration. The Fourier transform of δ and V_{IPMC} in the wireless and wired configurations is presented in Fig. 18, which shows that the spectrum of the propulsor tip deflection is largely monochromatic consistent with Figs. 16 and 17. As expected, the maximum tip deflection is attained close to the nominal actuation

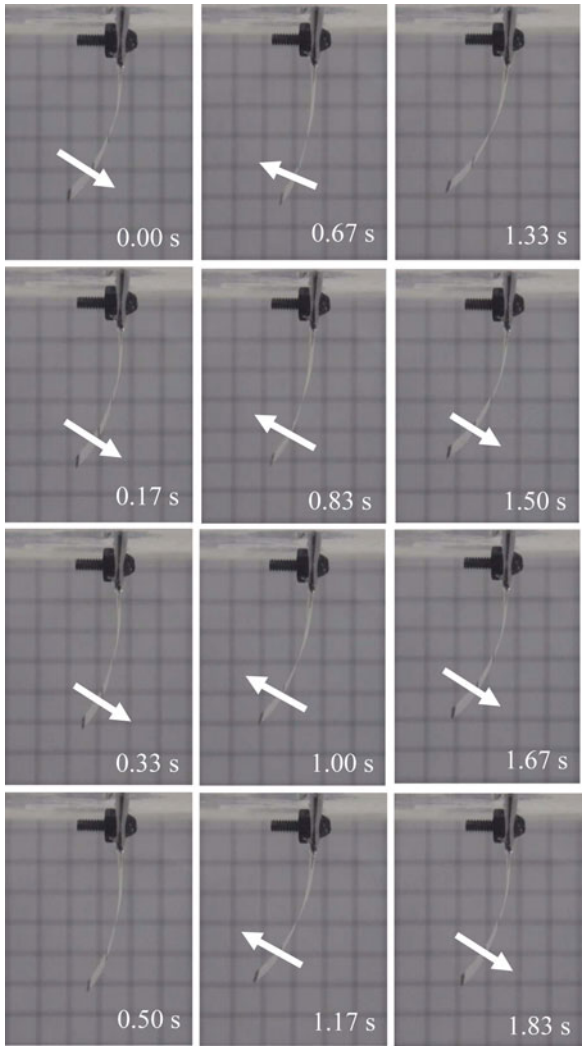


Fig. 16. Snapshots corresponding to approximately one period of IPMC actuation in the wirelessly powered configuration.

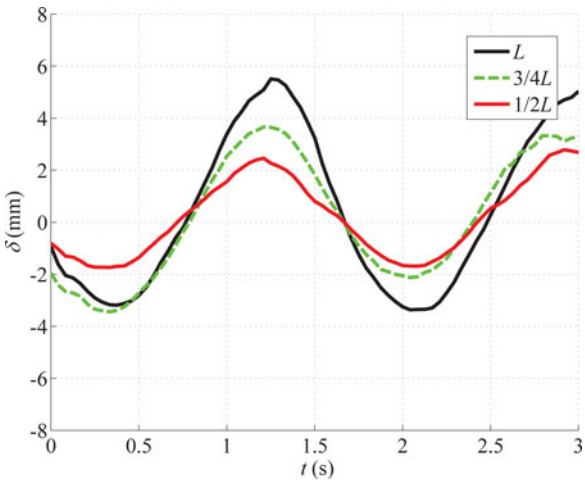


Fig. 17. Deflection of IPMC propulsor at half, three quarters, and full length in the wirelessly powered configuration.

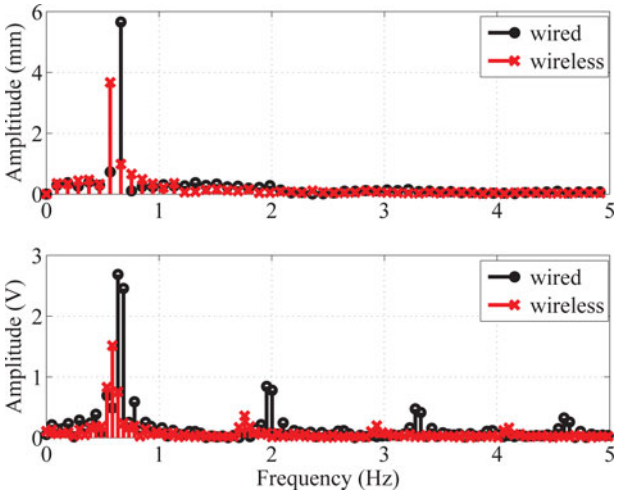


Fig. 18. Amplitude spectra of the voltage across the IPMC and the motion of the IPMC in wired and wireless configurations.

frequency of 720 mHz for both the wired and wireless cases and it is generally larger in the wired configuration. The same trends are also evident in the voltage signals in Fig. 18 which also displays considerable attenuation of higher frequency harmonics in the wireless configuration and slight differences between the frequency values at which the amplitude spectrum is maximal. Specifically, the amplitude spectrum of the voltage signal is maximized at 661 and 570 mHz for the wired and wireless cases, respectively. The higher harmonics observed in the IPMC voltage are due to the fact that the applied input is nominally a square wave. Such higher harmonics are not visible for the tip deflection due to the mechanical damping induced by the surrounding fluid as further illustrated in [60] and [63].

The effect of distortion in wireless powering is also evidenced by the time traces of the IPMC voltage V_{IPMC} in Fig. 19, which demonstrates that the well-defined square-wave shape in the wired configuration is degraded in the wireless case. This can be attributed to the alteration of the DC voltage at the power electronics from wireless transfer as visible in Fig. 19. Consequently, the power delivered to the IPMC in the wireless configuration is lower than in the wired configuration. In particular, by adapting the methods in Section III-B, we find that 530 mW is delivered to the IPMC in the wired case, while 235 mW is absorbed by the IPMC as it is wirelessly powered. We note that the measured power absorbed by the wirelessly powered IPMC is in good agreement with the earlier presented estimate of 198 mW, that is based on Fig. 10 and nominal conditions.

In order to qualitatively describe the fluid surrounding the vibrating IPMC propulsor, we perform a flow visualization experiment retaining the same coil arrangement as in the aforementioned analysis. A fine powder of cornstarch coats the surface of deionized water resting in an experimental tank of in-plane dimensions of 31 cm × 23 cm and height of 5.5 cm. A schematic of the setup is shown in Fig. 20. The IPMC propulsor is partially submerged so that a concave meniscus forms on it. The cornstarch cannot travel up the meniscus, and thus, a region surrounding the propulsor is void of cornstarch. As the IPMC

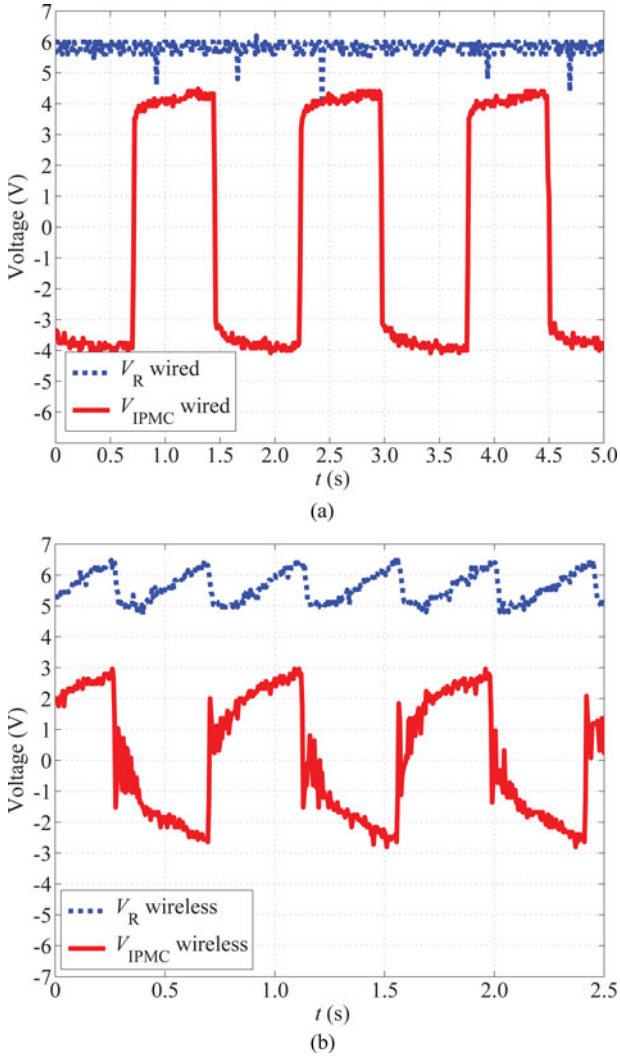


Fig. 19. DC voltage of the power electronics V_{R} and voltage across the IPMC V_{IPMC} in (a) wired and (b) wireless configurations.

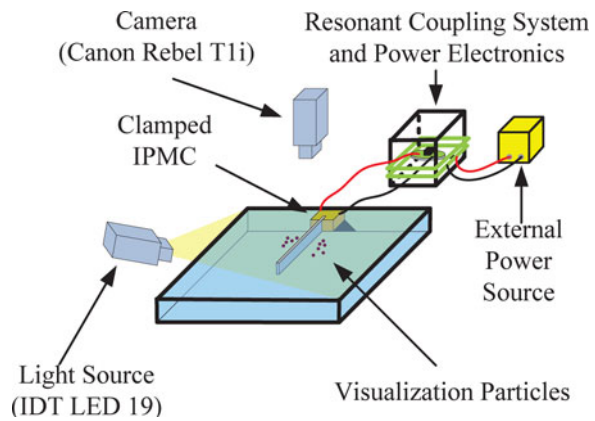


Fig. 20. Experimental setup for flow visualization of a vibrating IPMC in a quiescent fluid.

vibrates, the regions with and without cornstarch trace out the flow patterns seen in Fig. 21. Vortices are shed along two lines away from the tip of the propulsor carrying momentum associated with the thrust generated along the length of the element.

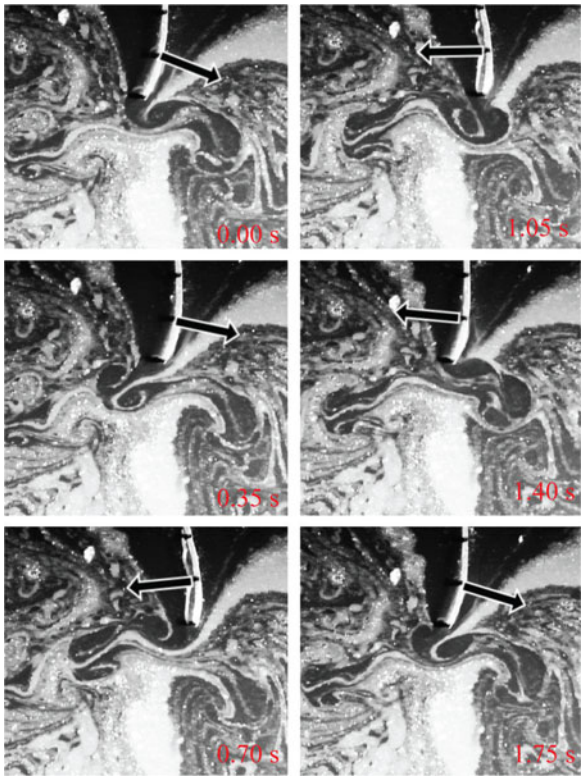


Fig. 21. Snapshots of the flow surrounding the IPMC after ten cycles of vibration.

Asymmetry in the curvature of the IPMC propulsor results in one vortex street being stretched transversely, while the other remains unstretched. This flow pattern is similar to the Japanese Fan Flow [64]. Fig. 21 also shows a sequence of six snapshots in time of the flow around the IPMC propulsor. When $t = 0.00\,\mathrm{s}$, the IPMC has reached its leftmost position at which time a jet ejects from the right side of the tip and forms a small vorticity pair. As the propulsor tip moves to the right, fluid in its path wraps around the tip of the propulsor to the left, as seen when $t = 0.35\,\mathrm{s}$ and $t = 0.70\,\mathrm{s}$. The vorticity pair is separated as the IPMC returns to the left, as seen when $t = 1.05\,\mathrm{s}$ and $t = 1.40\,\mathrm{s}$. Finally, $t = 1.75\,\mathrm{s}$ shows the ejection of another jet as the IPMC reaches its leftmost position. In other words, it is observed that fluid departs from the propulsor tip most predominantly at the two extremes of the oscillation and that the vortices emanating from the tip are deflected along preferential paths defining the jets. The flow visualization qualitatively suggests that the wirelessly powered IPMC generates sufficient fluid motion to be useful in an application as an active surface for hovering microswimmers.

An estimate of the propulsive capacity of the wirelessly powered IPMC can be obtained from planar PIV (see, for example, [38] and [37]). The experimental setup for the thrust measurement of the wirelessly powered IPMC is shown in Fig. 22. Herein, the IPMC is clamped between two aluminum contacts and its vibration amplitude and frequency are modulated by the external power source and power electronics to be consistent

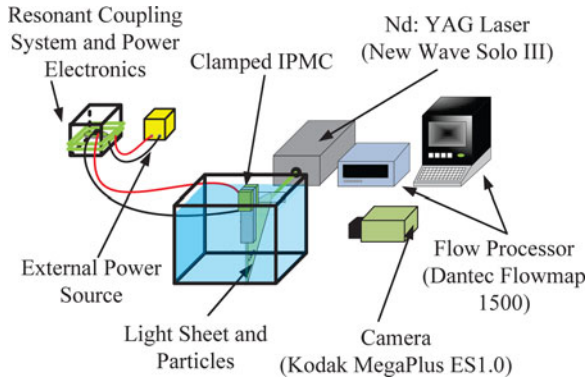


Fig. 22. Schematic representation of the PIV setup for the thrust determination of the wirelessly powered IPMC.

with previous motion tracking experiments of the wirelessly powered IPMC.

The IPMC propulsor is immersed in a 22 L water tank which is seeded with silver-coated hollow glass microspheres, from Potters Industries Inc., that are $17\,\mu\mathrm{m}$ in diameter. The particles are illuminated by a Nd:YAG laser from New Wave Solo III that emits $100\,\mathrm{mJ}$ per pulse. The laser beam is expanded using a cylindrical lens and focused by another lens into the water tank. The thickness of the resulting light plane is approximately $2\,\mathrm{mm}$ and it is positioned along the centerline of the vibrating IPMC. Images of the illuminated plane are captured via a Kodak MegaPlus ES 1.0 charge coupled device (CCD) camera with an AF Micro Nikkor $60\,\mathrm{mm}$ focal length lens. The captured field of view is approximately $100 \times 100\,\mathrm{mm}^2$. The PIV system performs a time-resolved analysis of the underwater vibration by sampling at $15\,\mathrm{Hz}$ over five cycles of oscillation where the time delay between image acquisitions is $25\,\mathrm{ms}$. The camera and laser synchronization as well as the image processing are performed by a Dantec Dynamics Flowmap 1500 system. Each acquired image is divided into a $32 \times 32\ \mathrm{pixel}^2$ interrogation region where adaptive correlation is used to resolve velocity vectors from the acquired images, see [37].

The force exerted by the IPMC on the surrounding fluid is determined by performing a control surface analysis on the 2-D measured velocity field shown in Fig. 23, see [38, eq. (5)]. We note that the 2-D nature of PIV measurements does not account for the volumetric flow rate of fluid leaving the control surface in the out-of-plane direction. Therefore, we compute the contribution of the out-of-plane flux to the measured force that is neglected when using a control surface as opposed to a control volume via [38, eq. (8)]. The dimensions of the control surface are chosen such that small changes in them do not yield large changes in the computed thrust.

When wirelessly powered, the IPMC vibrates at the frequency $f = 570\,\mathrm{mHz}$ with the maximum tip displacement $\hat{\delta} = 4.35\,\mathrm{mm}$ as determined in Figs. 17 and 18. Thus, the maximum tip speed can be approximated by $2\pi f\hat{\delta} = 16.4\,\mathrm{mm/s}$. We define the local Reynolds number as $Re_{\mathrm{f}} = 2\pi f\hat{\delta}L/\nu$ as suggested in [39]. Here, ν is the kinematic viscosity of water at room temperature and L is the free length of the IPMC and passive fin. The Reynolds number for the wirelessly powered

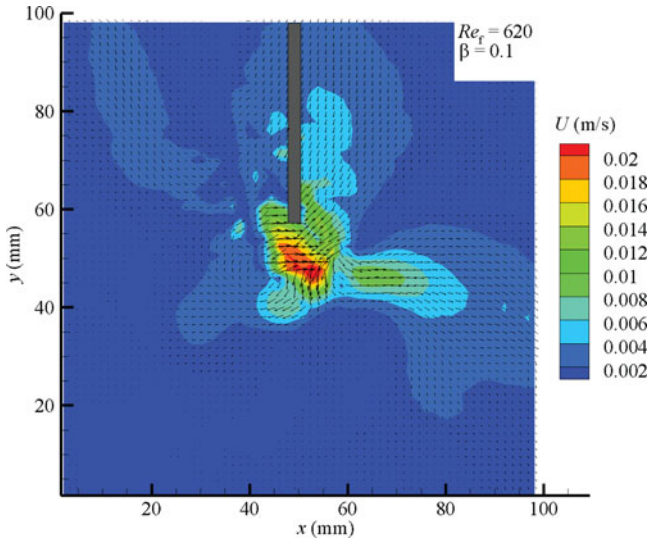


Fig. 23. Time-averaged velocity magnitude generated by a wirelessly powered IPMC propulsor. The solid black rectangle identifies the IPMC propulsor with exaggerated thickness.

IPMC is $Re_f = 620$, the relative tip displacement, $\beta = \hat{\delta}/L$, is 0.11, and the aspect ratio, $\lambda = w/L$, where w represents the width of IPMC passive fin, is $\lambda = 0.38$.

We compute the average thrust $\bar{\mathcal{T}}$ per unit width of the vibrating IPMC propulsor to be $0.46\,\mathrm{mN/m}$, while the contribution of the out-of-plane flux to the measured force is determined to be $0.058\,\mathrm{mN/m}$. The contribution of the out-of-plane flux to the measured force is $\sim 10\,\%$, which is comparable to findings reported in [38] that range from 10 to 20 %. The computed average thrust per unit width is compared with the power law fit of numerical data from [39], wherein it is determined that for $\beta = 0.1$, the average thrust per unit width is given by $\bar{\mathcal{T}} = 1.13 \times 10^{-9} Re_f^2$ which yields $\bar{\mathcal{T}} = 0.44\,\mathrm{mN/m}$. In addition, $\bar{\mathcal{T}}$ may also be compared with [38, Fig. 12] upon considering a different definition of the Reynolds number based on w instead of L, which would then be equal to 230. Therein, for a Reynolds number of 300, an aspect ratio of $\lambda = 0.38$, and a relative tip displacement of $\beta = 0.065$, a mean thrust per unit width of $\bar{\mathcal{T}} \simeq 0.1\,\mathrm{mN/m}$ is estimated while for an aspect ratio of $\lambda = 0.25$ and a relative tip displacement of $\beta = 0.164$, a mean thrust per unit width of $\bar{\mathcal{T}} \simeq 0.6\,\mathrm{mN/m}$ is found.

The proof-of-concept platform is capable of generating thrust comparable to that needed for underwater propulsion of miniature vehicles (see, for example, [30] and [31]). Further evidence for the practicality of the proposed system in wireless powering of microswimmers is procured from the direct measurement of the overall power absorbed by the resonant coupling system as it drives the power electronics connected to the IPMC propulsor. The measurement shows that the system absorbs $421\,\mathrm{mW}$, that is, the overall system efficiency, defined as the ratio between the IPMC consumption and the system power input, is 60 %, which is well in line with similar studies on wireless powering [41], [42] where efficiency on the order of 40–50 % is reported. We note that in the foreseen conceptual design in Fig. 1, the usable thrust by the robotic microswimmer is controlled by the projection of the propulsor vibration axis on the hovering direction which is a design parameter for hovering studies. Therefore, it is expected that fine parameter tuning will be needed to synthesize effective hovering robotic microswimmers. Parameters to be explored will entail buoyancy adjustment, IPMC propulsor initial curvature, and passive fins' material properties and geometry.

IV. Conclusion

In this paper, we have demonstrated a method to deliver power wirelessly to IPMCs. The components of the proof-of-concept platform were analyzed individually and en masse. The mutual inductances between the coils in the resonant coupling system were identified. We experimentally determined the power transferred to loads of incrementally varying shunting impedances. The electrical properties of the IPMC were described by a lumped circuit model and the associated model parameters were identified. In a flow visualization experiment, we observed the fluid motion in the wake of a wirelessly powered IPMC. Through image analysis, we analyzed the IPMC vibration and thus estimated its thrust production.

The platform shows IPMC performance largely comparable to IPMCs powered by a local power system and is thus expected to be an ideal candidate to enable experimental research in fluid mechanics. More specifically, the platform is expected to act as a centimeter-scale laboratory test bed for investigating fundamental problems in free locomotion, such as hovering. Future work will focus on transitioning the current platform into the realization of an actual microswimmer and will explore embedding parts of the receiving coil pair into the IPMC itself by following emerging fabrication methods, such as those reported in [65] and [66].

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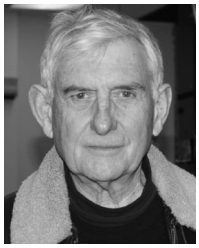
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